

BHAVYA RAKESHKUMAR PATEL

bhavyarpatel22@gmail.com || 438-833-9398 || Montreal, Quebec, Canada

www.linkedin.com/in/bhavyapatel1000

HIGHLIGHT OF QUALIFICATIONS

- Graduate student in Applied Computer Science at Concordia University, specializing in Artificial Intelligence, Machine Learning, and Data Science.
- Over 2 years of professional experience in AI/ML engineering, data-driven development, and model deployment across academic and industry projects.
- Strong proficiency in Python, TensorFlow, Scikit-learn, and LangChain, with practical expertise in LLMs, NLP, and transformer-based architectures.
- Hands-on experience integrating AI and cloud technologies using AWS Bedrock, Azure ML, and Streamlit to build scalable intelligent systems.
- Demonstrated leadership and teaching ability as a Teaching Assistant, coupled with strong communication and cross-functional collaboration skills.

EDUCATION

Master of Applied Computer Science (MAPCompSc)

Concordia University, Montreal, QC, Canada

Sept 2024 - Aug 2026

GPA: 3.33/4

Bachelor of Technology in Computer Engineering

Charotar University of Science and Technology (CHARUSAT), Changa, India

Jun 2020 - May 2024

CGPA: 9.13/10

SKILLS

- Technical Skills:** Python, C, C++, Java, Shell Scripting, SQL, Django, Flutter, REST API Integration, UI/UX Design, Web Scraping, Data Analytics
- AI & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, LangChain, Keras, Hugging Face, Natural Language Processing (NLP), Vector Databases, Prompt Engineering, LLM Frameworks
- Cloud & Tools:** AWS Bedrock, AWS SageMaker, Azure ML, Linux, Git, GitHub, Jira, CI/CD

WORK EXPERIENCE

Teaching Assistant

Concordia University, Montreal, Canada

Sep 2025 – Present

- Guided 20+ students in mastering Distributed Systems concepts such as Java RMI, CORBA, and SOAP-based Web Services through engaging lab demonstrations and personalized support.
- Facilitated weekly tutorials and 5+ lab sessions, aligning course content with practical exercises to maximize student understanding and participation.
- Evaluated 5+ assignments and final projects each term, ensuring consistent grading standards and delivering feedback within 5 days to foster continuous improvement.

Data Science Intern

Blue Data Consulting & IT Services Pvt. Ltd., Surat, India

Dec 2023 – Jun 2024

- Devised an AI-driven workflow converting 50+ PowerPoint Presentations into video courses using LLMs, RAG, and Streamlit, increasing automation efficiency by ~40%.
- Benchmarked 4 LLMs (GPT-3.5, Opus, Sonnet, Gemini), achieving 25% higher script accuracy and 35% faster generation with GPT-3.5-Turbo.
- Produced 10+ AI-based course modules (LangChain, Flowise AI, Synapse CoR, Prompt Engineering) in collaboration with a 5-member team, boosting content scalability by 3×.

Machine Learning Research Student

CHARUSAT University, Changa, India

Mar 2022 - Jun 2023

- Extracted and prepared 13K+ video frames from the UCF-101 Sports Action Dataset to train deep learning models for multi-sport activity recognition.
- Optimized the Vision Transformer (ViT) model, boosting classification accuracy from 84% to 94% through advanced hyperparameter tuning and data preprocessing.
- Investigated model performance across varying hyperparameter configurations and dataset augmentations, enhancing training stability and reducing validation loss by ~9%.

Flutter Development Trainee
Inspire Cyber Security, Surat, India

May - Oct 2022

- Prototyped 3+ mobile applications in Adobe XD, converting wireframes into responsive Flutter interfaces that accelerated iteration cycles by 50%.
- Constructed a Notes App using SQLite for offline storage, implementing CRUD functionality with zero data loss during local syncing.
- Delivered a client project — “Palm Box Cricket Booking System” — enabling real-time slot reservations and increasing user engagement by ~40% through an intuitive Flutter UI/UX.

Python Developer Intern
Jemistry Info Solutions LLP, Surat, India

May 2022 - Jul 2022

- Supervised a 4-member intern team to create a PC health monitoring tool using Python’s Socket & OS modules, automating real-time system reporting to server IPs.
- Implemented secure authentication in Django with PostgreSQL, reducing login latency by ~30% and ensuring robust user management.
- Reengineered backend modules to enhance efficiency, lowering system resource consumption by 40% on low-end devices while enabling enterprise-grade control features.

MAJOR PROJECTS

Multilingual Video Script Generation Web App

- Formulated an AI pipeline that transformed PowerPoint slides into narrated videos using GPT models and TTS APIs.
- Integrated automated script extraction, multi-language selection (5+ languages), and animation into a single seamless workflow.
- Accelerated video production by ~70%, embedding generated scripts into slide notes and exporting finalized outputs as MP4.

Museum View Detection- AI/ML

- Constructed baseline classifiers (Decision Tree, Random Forest, XGBoost) to distinguish indoor vs. outdoor museum imagery.
- Developed and fine-tuned a PyTorch CNN on the MIT Places dataset (10,000 images) using advanced data augmentation and hyperparameter optimization.
- Attained 91.5% accuracy and 91.4% F1-score, surpassing baseline models by 16–22% in classification performance.

PUBLICATIONS

Vision Transformer in Sport Action: Recognizing Athletic Activities across Varied Sporting Domain

Jun 2024

- Co-authored a research paper introducing a novel Vision Transformer (ViT)–based approach for sports action recognition, enhancing classification accuracy across multiple athletic categories.
- Validated the model on a dataset of 10,000+ frames, achieving substantial performance improvements over traditional CNN architectures.
- Presented findings at the International Conference on Data Science Applications (ICDSA 2024), India, and published in the Springer Book Series.
- DOI: https://link.springer.com/chapter/10.1007/978-981-96-2179-8_35

CO-CURRICULAR ACTIVITIES

Professional Service Director

Sep 2022 – Aug 2023

Rotaract Club of CHARUSAT, CHARUSAT University, Changa, India

- Coordinated and led 10+ social, technical, and cultural events, impacting over 500 students and community members.
- Directed team operations and event logistics for 10+ initiatives, improving coordination efficiency by 25% and strengthening leadership and conflict-resolution outcomes.

Event Volunteer

Jul 2022

Hackathon India, CHARUSAT, Changa, India

- Actively participated in coding challenges and real-time product design sprints across 2+ national-level hackathons.
- Volunteered in operations and mentoring coordination, helping over 200 participants during registration, ideation, and demo rounds.

CERTIFICATIONS

- **Fundamentals of Digital Image and Video Processing (Coursera – Northwestern University):** Learned core concepts in signal processing, motion estimation, and video compression using MATLAB and Python.
- **Architecting Smart IoT Devices (Coursera – EIT Digital):** Gained practical knowledge in IoT system design, including sensor integration, edge computing, and secure communication protocols.
- **Flutter Development with UI/UX (Inspire Cyber Security):** Trained in Flutter and Dart to build cross-platform apps with Firebase backend and intuitive UI/UX.